

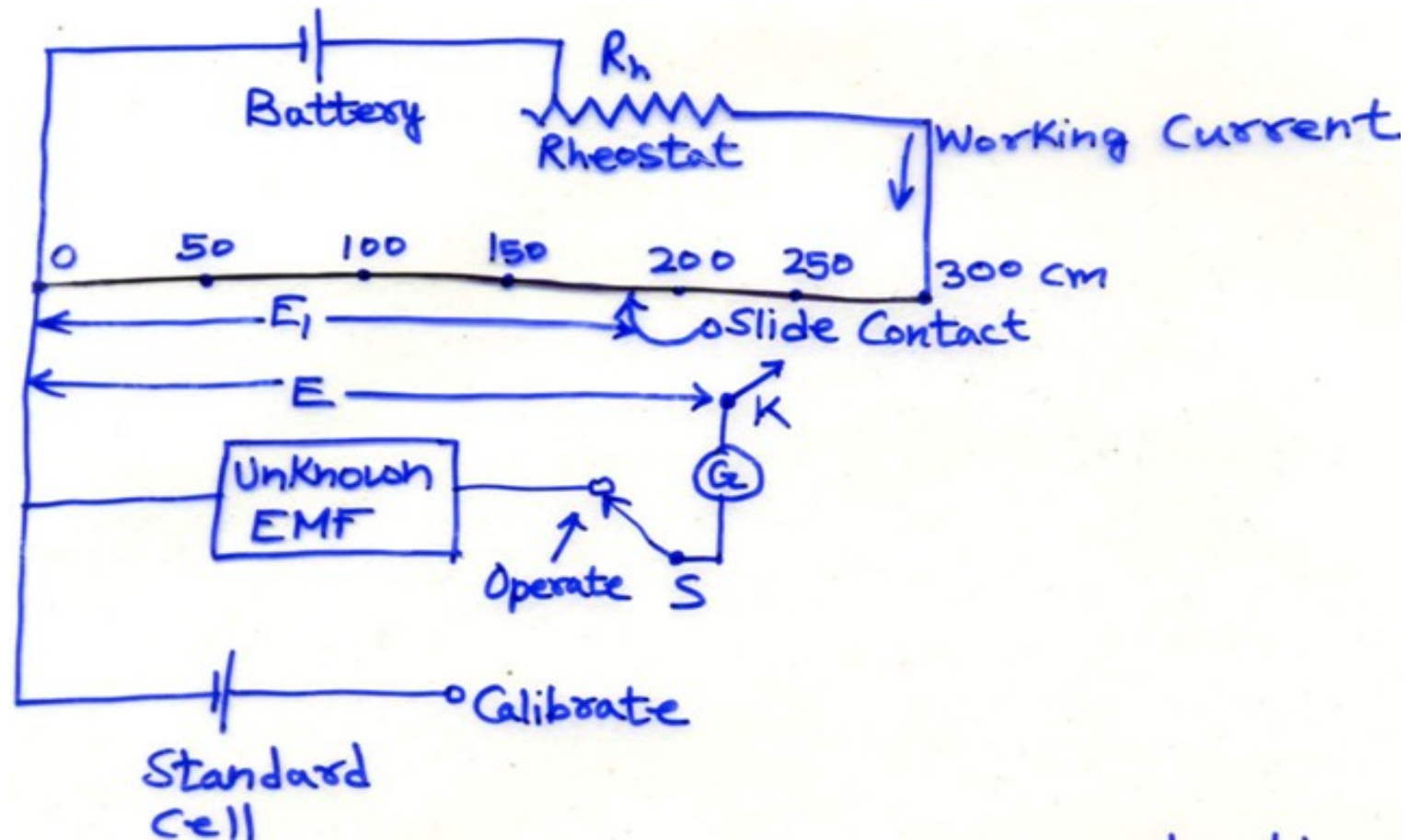
Potentiometers

Introduction

- Instrument used to measure unknown voltage by comparing with a known.
- Very accurate.
- No power consumed in the circuit containing the unknown emf at balance.
- Voltage determination is independent of source resistance.

DC Potentiometer

Basic Slide Wire Potentiometer



Standardisation

- The process of adjusting the working current so as to match the voltage drop across a portion of sliding wire against a standard reference.

Slide wire length = 200 cm and resistance = 200 Ω .

Emf of standard cell = 1.0186 V.

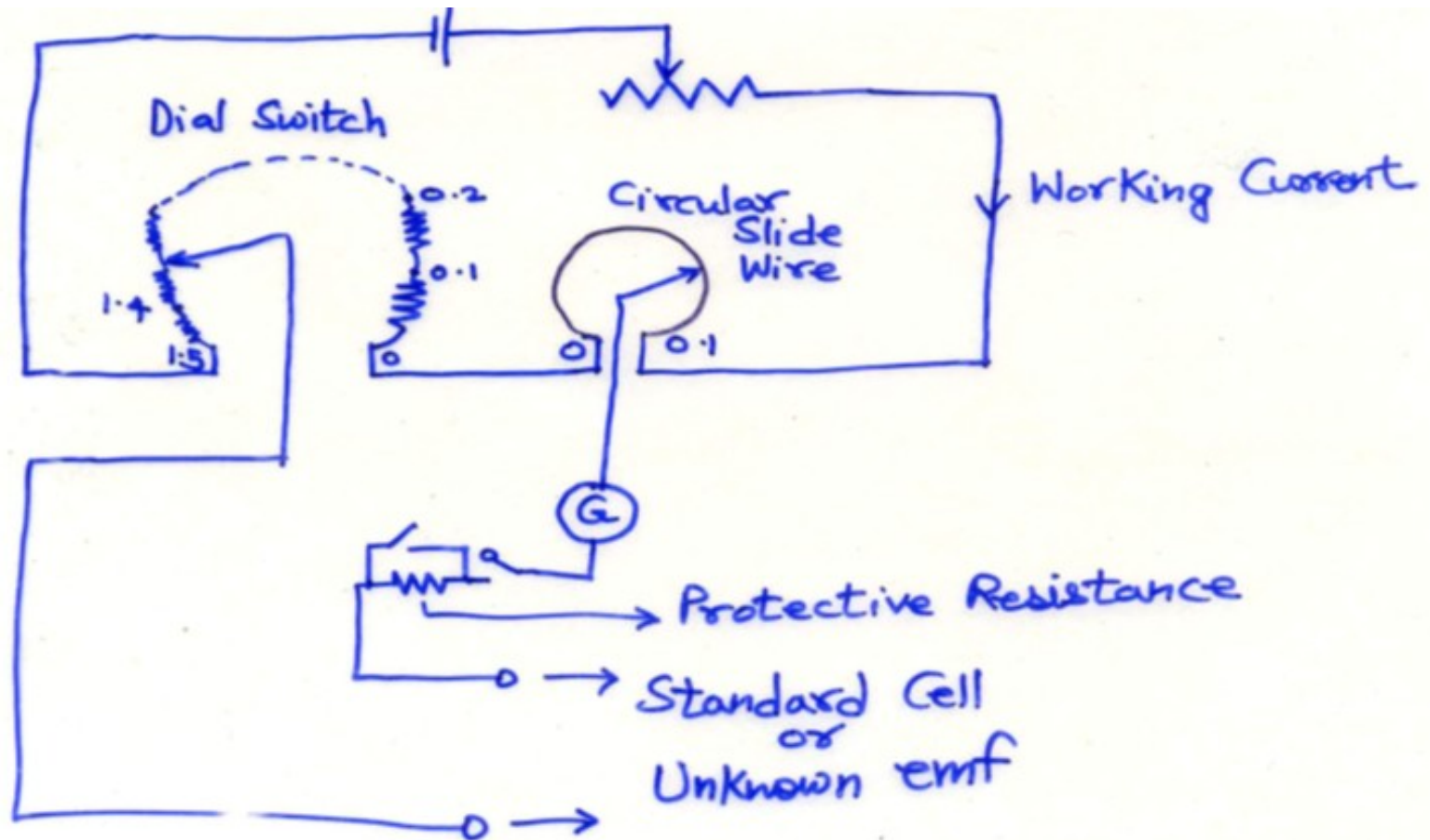
Sliding contact placed at 101.86 cm mark and current adjusted till zero deflection.

$$\Rightarrow \text{Working current} = \frac{1.0186}{101.86} = 10 \text{ mA}$$

Constructional Details

- All the resistors in a potentiometer except the slide wire are made of manganin.
- Manganin has high stability, low temperature coefficient and freedom from thermoelectric effects against copper.
- Slide wire is made from platinum-silver alloy.
- Sliding contact are of a copper-gold-silver alloy.

Crompton's Potentiometer



Standardisation of Crompton's Potentiometer

One dial with 15 steps. Dial resistors have a value of $10\ \Omega$ each.

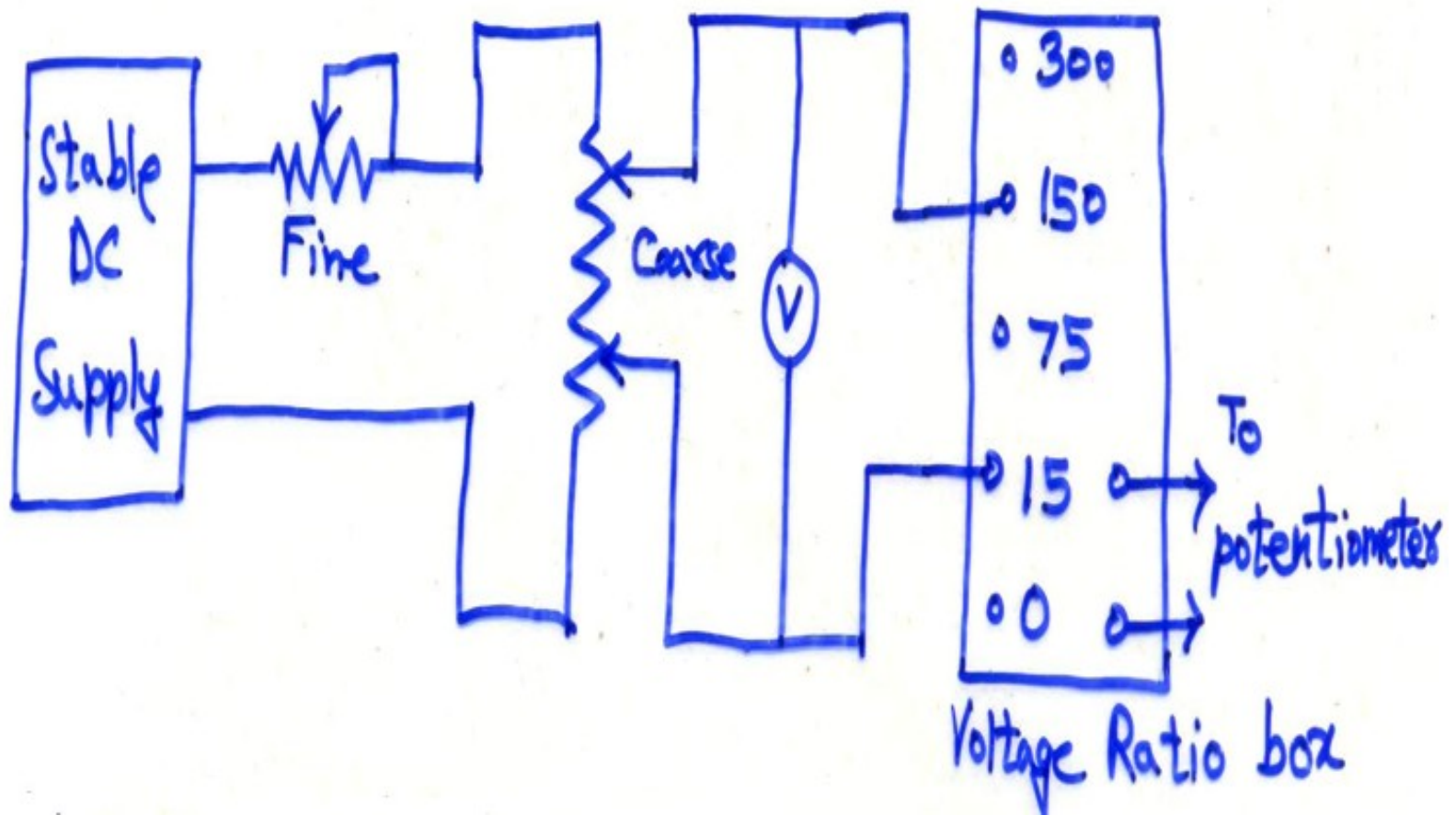
$$\text{Total resistance} = 15 \times 10 + 10 = 160\ \Omega$$

If working current = $10\ \text{mA}$, each step of dial switch will correspond to $0.1\ \text{V}$

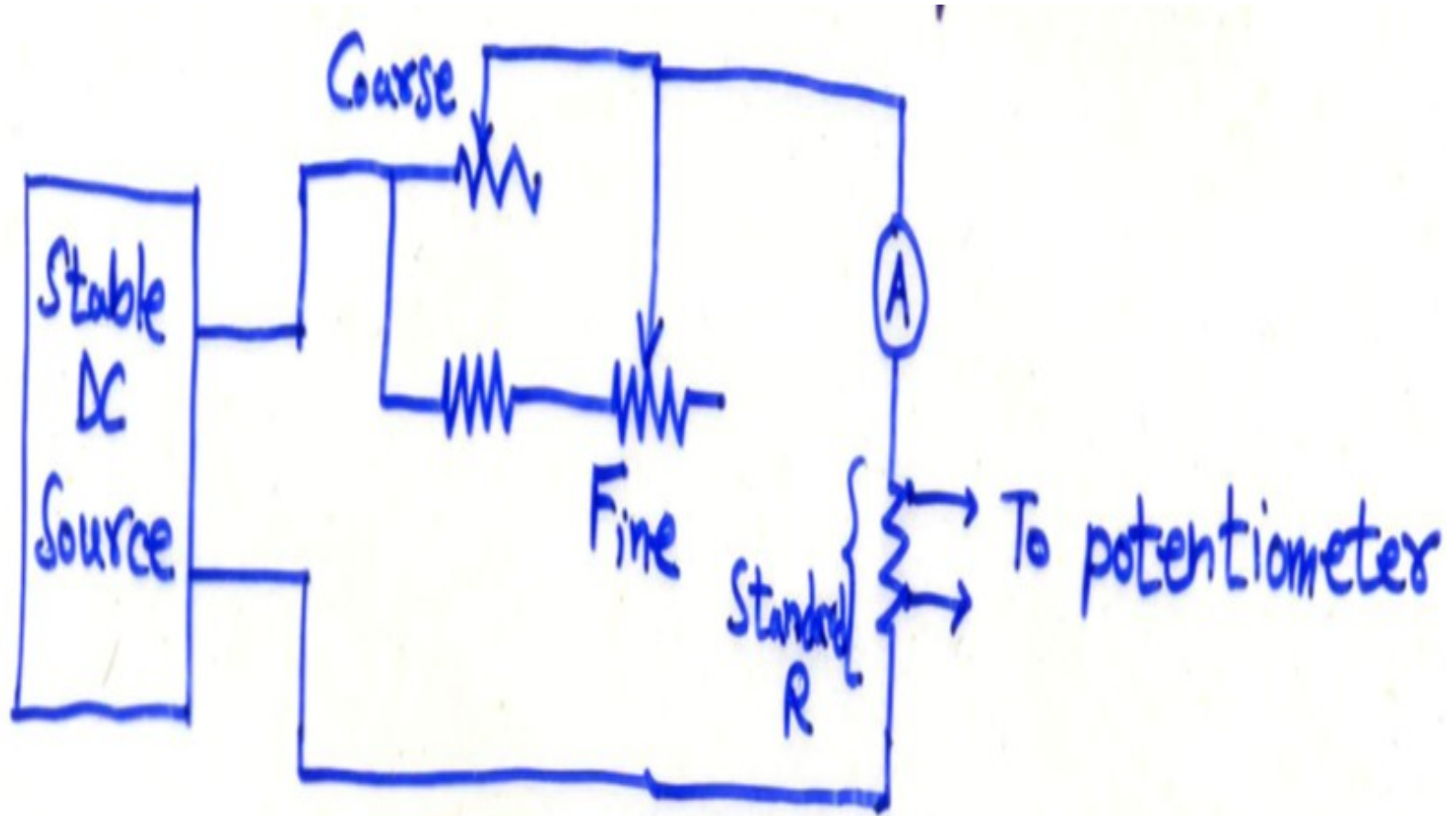
If there are 200 divisions, each div. of slide wire corresponds to $0.0005\ \text{V}$

Applications

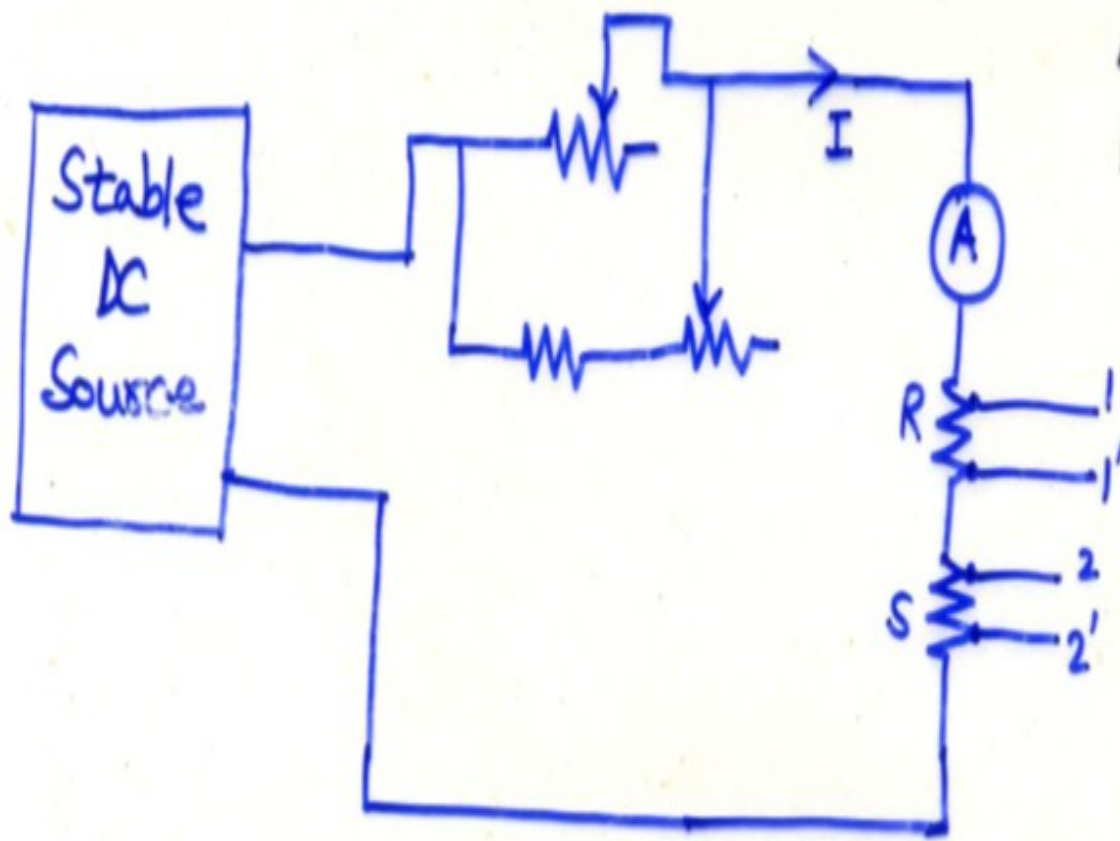
Voltmeter Calibration



Ammeter Calibration



Resistance Measurement



At 11'
Reading is V_R
At 22'
Reading is V_S

To
potentiometer

$$\frac{V_S}{S} = \frac{V_R}{R}$$

AC Potentiometer

Types

(1) Polar type

Voltage read in the form V/θ

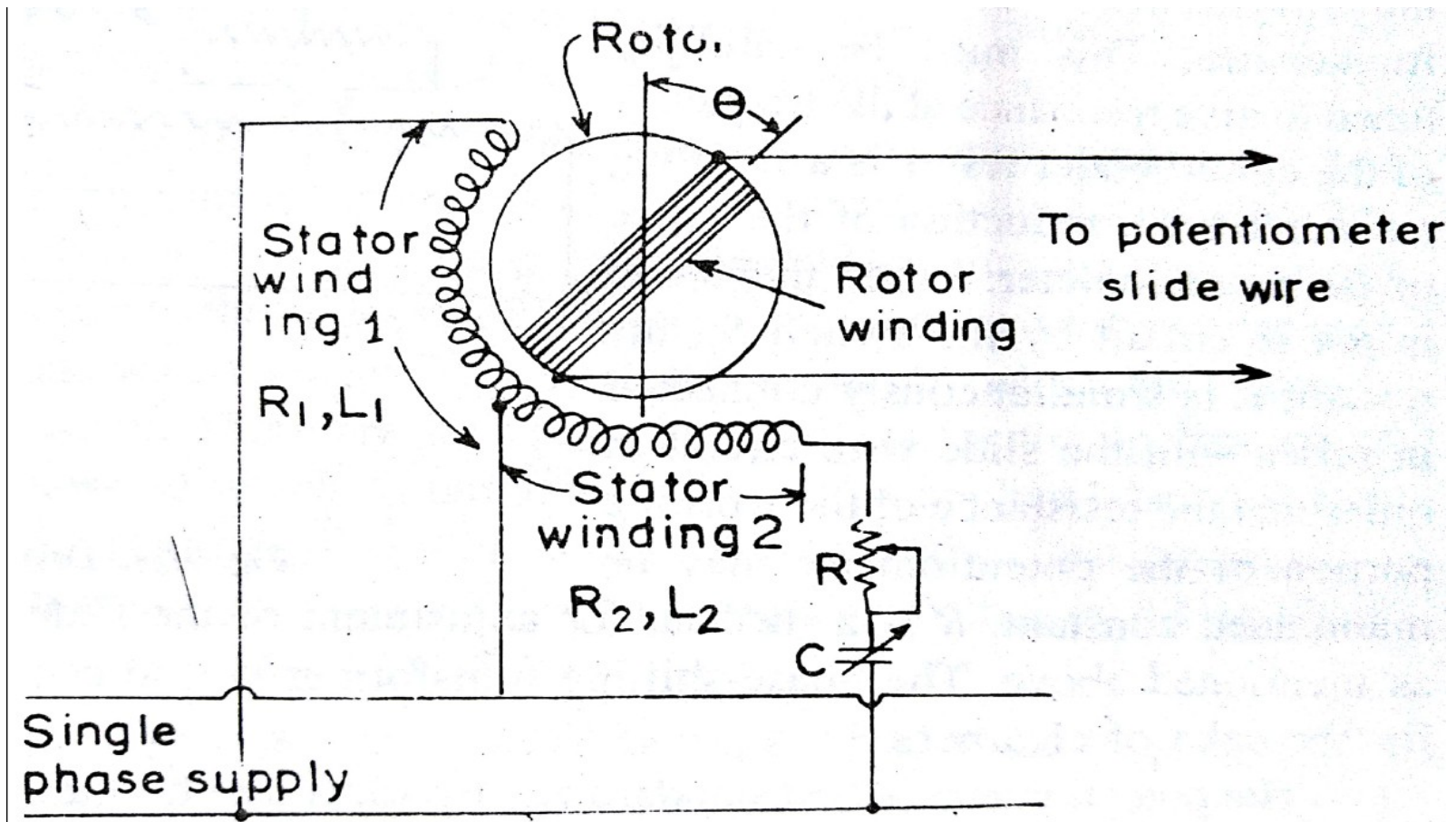
(2) Co-ordinate Type

$$V = (V_1^2 + V_2^2)^{1/2}$$

$$\theta = \tan^{-1} \left(\frac{V_2}{V_1} \right)$$

Polar Type

Drysdale phase shifting transformer



- Rotor can be adjusted through any required angle.
- Phase displacement of rotor emf = angle through which the rotor has been moved through its zero position.
- Scale and pointer indicates the angle.

Emf induced due to stator winding

$$e_1 = KI \sin \omega t \cos \theta$$

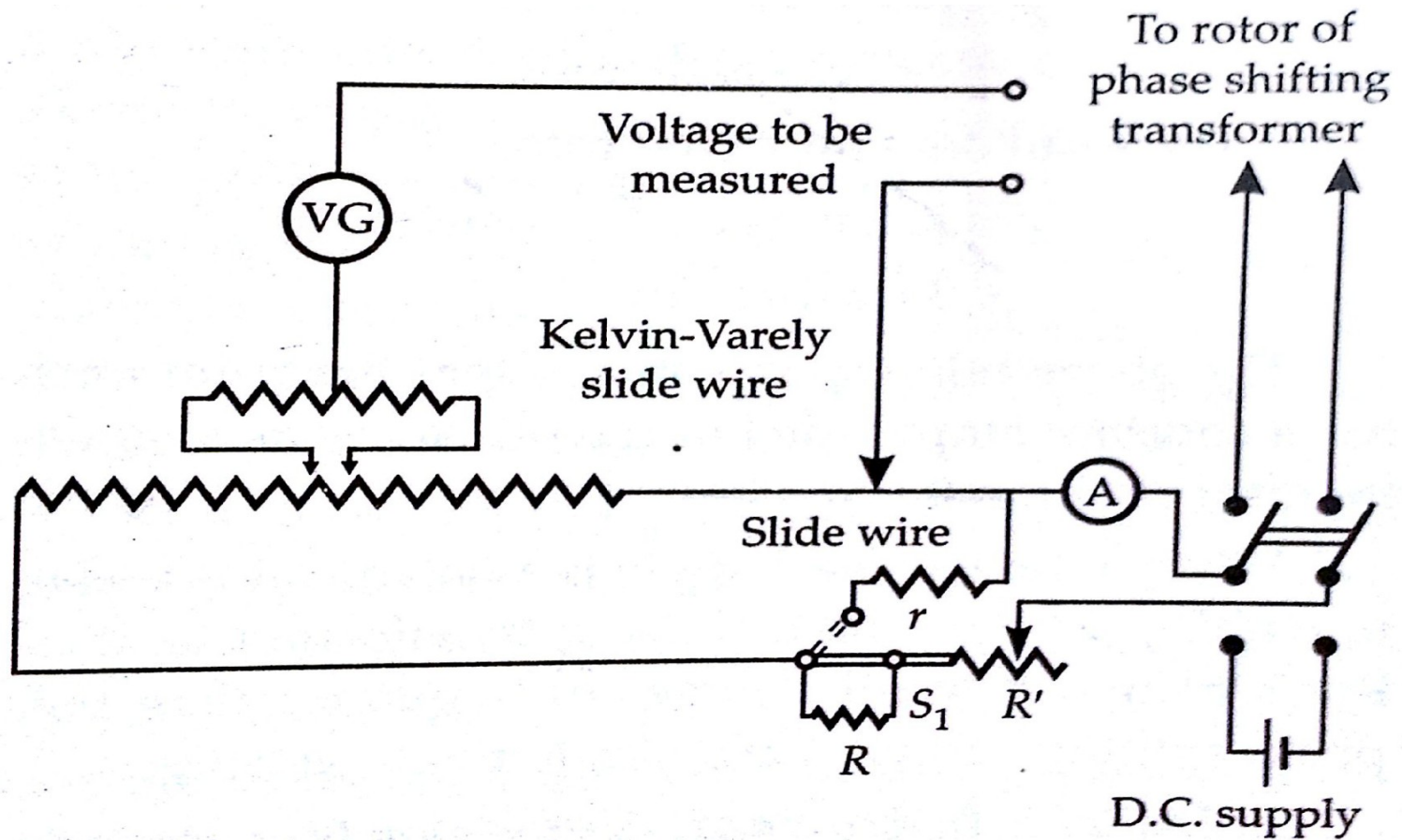
& that due to stator winding

$$e_2 = KI \sin (\omega t + 90^\circ) \cos (\theta + 90^\circ)$$

$$\Rightarrow e_2 = -KI \cos \omega t \sin \theta$$

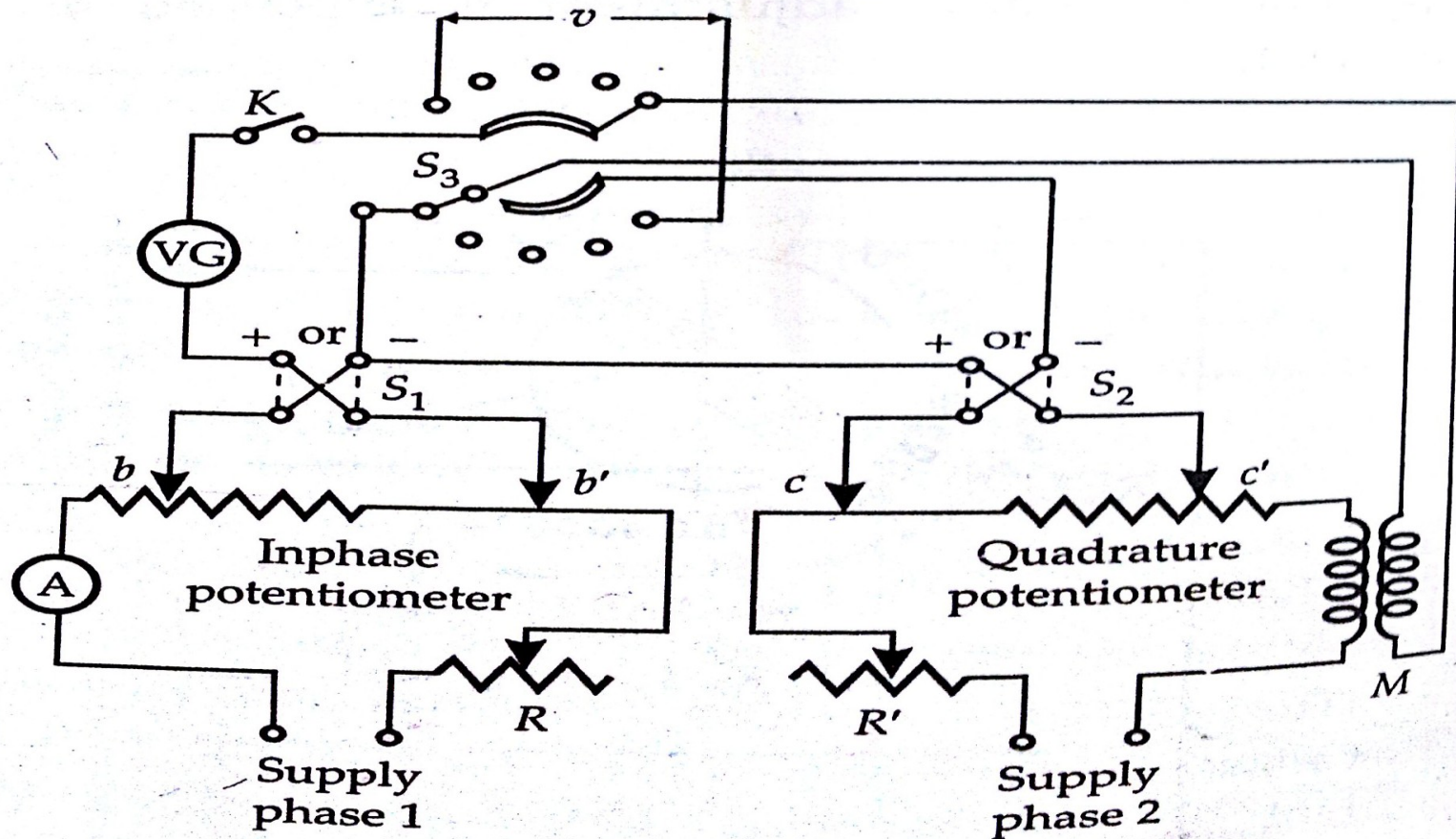
Resultant emf in rotor (e) = $KI \sin (\omega t - \theta)$

Drysdale Potentiometer

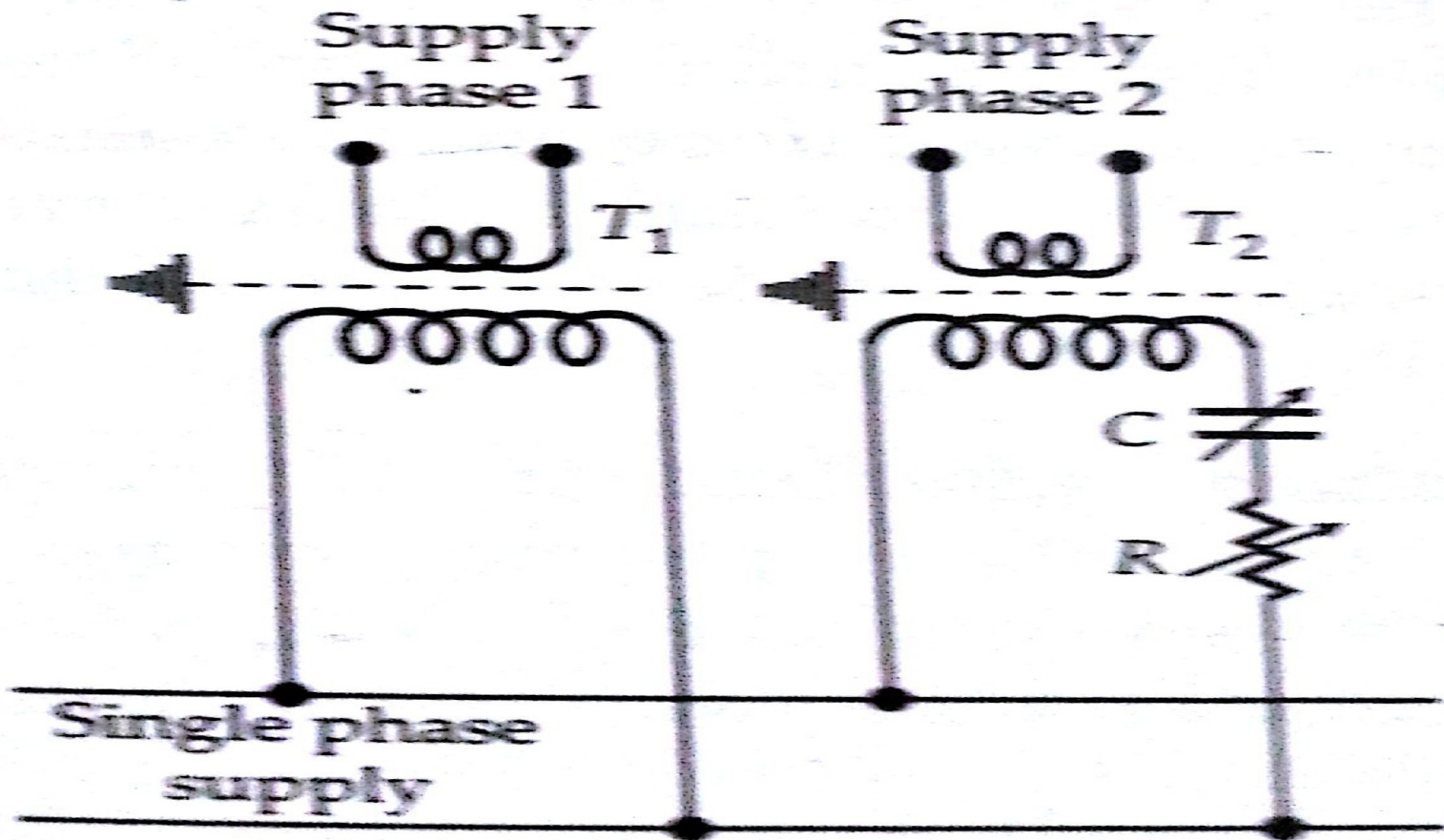


Co-ordinate Type

Gall Tinsley Potentiometer



Phase Splitting Circuit



DPDT Switch

